

JIFENG WU

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EDUCATION

CORNELL UNIVERSITY

Ithaca, NY

PhD Candidate

AUGUST 2025 - EXPECTED JUNE 2030

UNIVERSITY OF BRITISH COLUMBIA

Vancouver, Canada

MSc in Computer Science

NOVEMBER 2024

- Advisor: Caroline Lemieux
- GPA 4.0, Coursework: Software Engineering, Automated Testing, Type Systems, Deep Learning

WUHAN UNIVERSITY

Wuhan, China

BEng in Software Engineering, Outstanding Graduate

JUNE 2022

- Advisor: Qingan Li
- GPA 3.93, Coursework: Operating Systems, Database Systems, Software Engineering, Embedded Software Design, Computer Systems: A Programmer's Perspective, Compilation Technology and Application

PUBLICATIONS

QuAC: Quick Attribute-Centric Type Inference for Python

Jifeng Wu, Caroline Lemieux

Proceedings of the ACM on Programming Languages, Issue OOPSLA, 2024

Effective Stack Wear Leveling for NVM

Jifeng Wu, Wei Li, Libing Wu, Mengting Yuan, Chun Jason Xue, Jingling Xue, Qingan Li

IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, 2023

RESEARCH EXPERIENCE

CORNELL UNIVERSITY

Ithaca, NY

Porting Deep Learning Applications to Neuron & Workflow Automation

SEPTEMBER 2025 - PRESENT

- Refactoring deep learning applications (e.g., GPT-OSS, YOLO) to enable efficient model inference on Amazon Neuron hardware.
- Built *safetensors-layer-grabber* (available on PyPI), a utility for targeted extraction of model weights from HuggingFace SafeTensors files.
- Developing and publishing multiple PyPI packages to transform Python bytecode into SSA-form intermediate representations for static analysis.
- Developing a dynamic gray-box methodology for systematic analysis and simplification of PyTorch models, enabling the extraction of interpretable module hierarchies and synthesis of weight-compatible classes.

UNIVERSITY OF BRITISH COLUMBIA

Vancouver, Canada

QuAC: Quick Attribute-Centric Type Inference for Python

JANUARY 2023 - AUGUST 2024

- Implemented QuAC, a novel Python type inference tool using attribute sets and information retrieval.
- Demonstrated high coverage and accuracy, outperforming non-LLM baselines, especially in predicting container type parameters and non-built-in types, while significantly reducing run times.
- Compared to LLM-based methods, QuAC is nearly two orders of magnitude faster while achieving greater consistency in its container type parameter predictions.
- GitHub: github.com/jifengwu2k/quac

WUHAN UNIVERSITY**Wuhan, China****Effective Stack Wear Leveling for NVM****AUGUST 2021 - AUGUST 2022**

- Proposed Loop2Recursion, a stack wear leveling technique implemented as an LLVM pass for increasing the lifespan of NVM with limited write durability by converting wear-heavy loops into recursive functions.
- Outperforms state-of-the-art methods by significantly improving stack wear leveling and reducing performance overhead.
- GitHub: github.com/jifengwu2k/loop2recursion

INDIANA UNIVERSITY BLOOMINGTON**Bloomington, Indiana****Automated Theorem Proving****SEPTEMBER 2024 - MARCH 2025**

- Conducted research under Prof. Sam Tobin-Hochstadt on Cheesecake, an inductive theorem prover leveraging SMT solvers and ACL2 heuristics. Investigated AI-based lemma conjecturing.

WUHAN UNIVERSITY**Wuhan, China****Community Detection Using Social Networks and Trajectories****SEPTEMBER 2019 - AUGUST 2021**

- Explored using trajectories to identify spatiotemporally cohesive user groups in large social networks.
- Developed a linear-time spatiotemporal trajectory similarity algorithm and community detection algorithm.
- Comprehensive evaluations on two datasets demonstrated the effectiveness and efficiency of these algorithms.

TEACHING EXPERIENCE

CORNELL UNIVERSITY**Ithaca, NY****Teaching Assistant, ECE 2400: Computer Systems Programming****JANUARY 2026 - APRIL 2026**

- Supported students in computer systems programming and C/C++ development through technical guidance, debugging help, and course support.
- Developed assignment infrastructure, including repository management tools, assignment releases, grading workflows, final grading scripts, and GitHub Actions self-checking.
- Delivered section presentations on Unix tooling, C++ environment setup, and C++ profiling.

UNIVERSITY OF BRITISH COLUMBIA**Vancouver, Canada****Graduate Teaching Assistant, CPSC 410: Advanced Software Engineering****SEPTEMBER 2022 - DECEMBER 2022**

- Supported students on static and dynamic program analysis course projects.
- Provided technical feedback on project design, implementation, and analysis strategies.
- Helped students understand advanced programming language and software engineering concepts through individualized guidance.

OPEN-SOURCE CONTRIBUTIONS

CHATREPL**Python 2+**

- An agent for OpenAI-compatible Chat Completions APIs.
- Features 4 tools: read, write, edit, shell.
- Supports streaming responses, multiline input via \$EDITOR, and file integration.
- PyPI: pypi.org/project/chatrepl

COPY-ON-WRITE LIST**Python 2+**

- An immutable, hashable, copy-on-write list-like data structure implementing the Sequence protocol.
- Supports O(1) indexing/slicing/appending-at-end and structural sharing.
- PyPI: pypi.org/project/cowlist

AVL ORDER STATISTIC SET

C++11

- C++11 header-only self-balancing AVL set with $O(\log n)$ order-statistics/range queries.
- API-compatible with `std::set`, portable replacement for GNU PBDS.
- GitHub: github.com/jifengwu2k/avl-order-statistic-set

INTERACTIVE DEPENDENCY RESOLVER

Python 2+

- A tool to interactively resolve Python wheel dependencies for debugging "dependency hell" scenarios.
- Supports platform/wheel compatibility, as well as dependency graph traversal.
- PyPI: pypi.org/project/interactive-dependency-resolver

SDIAL

Python 2+

- Command-line speed dial tool for quickly launching long commands by number.
- Supports multiline command editing via `$EDITOR` and passing arguments to commands.
- OS and shell agnostic, plain-text, backup-friendly storage.
- PyPI: pypi.org/project/sdial

FRAMEPERFECT

Python 2+

- Lightweight GUI application for precise frame-by-frame video analysis.
- PyPI: pypi.org/project/frameperfect

SKILLS

- Languages: Python (Expert), C/C++ (Advanced), Bash.
- Frameworks: PyTorch, NetworkX, PyQt/PySide, LLVM, MLIR, Intel Pin, Git, Docker.
- Domains: Program Analysis, Formal Methods, Compilers, Automated Reasoning.

AWARDS

- 3x Outstanding Student Scholarship (Top 5%); 2x Advanced Individual in Social Work
- China Software Cup (Second Prize, 2020); Service Outsourcing Innovation and Entrepreneurship Competition for Chinese College Students (Third Prize, 2021)